

Wireless & Mobile Communication

Unit II — Exam Preparation: 2-Mark & 5-Mark Questions with Answers

Syllabus: FDMA, TDMA, CDMA, comparison of multiple-access techniques, spectral efficiency calculations, wireless network planning, link budget and power spectrum calculations.

PART A — 2-MARK QUESTIONS WITH ANSWERS

Multiple Access Fundamentals

Q1. What is multiple access in wireless communication?

Answer: Multiple access is a technique that allows multiple users to share a common communication medium or radio spectrum while keeping their signals distinguishable.

Q2. What is FDMA? State its basic principle.

Answer: FDMA (Frequency Division Multiple Access) divides the available frequency spectrum into separate frequency bands and assigns a different frequency band to each user. Users can transmit continuously on their assigned bands.

Q3. What is TDMA? State its basic principle.

Answer: TDMA (Time Division Multiple Access) allows multiple users to share the same frequency channel by assigning different time slots to different users.

Q4. What is CDMA? State its basic principle.

Answer: CDMA (Code Division Multiple Access) allows multiple users to transmit at the same time and over the same frequency band by assigning different spreading codes to the users.

Q5. What is a guard band and why is it used in FDMA?

Answer: A guard band is a small unused frequency range placed between adjacent FDMA channels. It reduces adjacent-channel interference caused by imperfect filtering and signal leakage.

Q6. What is guard time and why is it used in TDMA?

Answer: Guard time is a small unused time interval between TDMA time slots. It prevents overlap between adjacent transmissions due to synchronization errors, propagation delay and timing uncertainties.

Q7. What is a spreading code in CDMA?

Answer: A spreading code is a sequence assigned to a user in CDMA. It spreads the user's signal over a wider bandwidth and allows the receiver to identify and recover that user's signal.

Q8. What is a chip in CDMA?

Answer: A chip is a small element of a CDMA spreading sequence. A single information bit may be represented by several chips, so the chip rate is normally higher than the data bit rate.

Q9. What is an orthogonal code?

Answer: Orthogonal codes are codes whose cross-correlation is ideally zero. They allow multiple CDMA users to share the same time and frequency while being separated at the receiver.

Q10. What is the near-far problem in CDMA?

Answer: The near-far problem occurs when a strong signal from a nearby user overwhelms a weak signal from a distant user at the same base station. Accurate power control is used to reduce this problem.

Q11. Why is power control important in CDMA?

Answer: Power control adjusts users' transmission powers so that no user's signal is unnecessarily strong or weak at the receiver. It reduces interference and helps overcome the near-far problem.

Q12. Define processing gain in CDMA and write its formula.

Answer: Processing gain is the ratio of spread bandwidth (or chip rate) to the original data rate. $G_p = W/R = \text{Chip Rate}/\text{Data Rate}$.

Spectral Efficiency**Q1. Define spectral efficiency and state its unit.**

Answer: Spectral efficiency is the amount of information transmitted per unit bandwidth. Its unit is bits/s/Hz (bps/Hz).

Q2. Write the basic formula for spectral efficiency.

Answer: The basic formula is: $\eta = R/B$, where R is data rate in bits/s and B is bandwidth in Hz.

Q3. What is meant by bits/s/Hz?

Answer: Bits/s/Hz means the number of bits transmitted per second for every 1 Hz of bandwidth used.

Q4. Why is spectral efficiency important?

Answer: Radio spectrum is limited. Higher spectral efficiency allows more information or users to be supported using the same amount of bandwidth.

Q5. How does guard-band overhead affect FDMA spectral efficiency?

Answer: Guard bands consume part of the available spectrum without carrying user information. Therefore, guard bands reduce the practical spectral efficiency of FDMA.

Q6. How does guard time affect TDMA efficiency?

Answer: Guard time does not carry user data. It reduces the fraction of each frame used for useful transmission and therefore lowers practical TDMA efficiency.

Q7. Write the formula for CDMA processing gain.

Answer: $G_p = W/R$, where W is spread bandwidth and R is data rate. Equivalently, $G_p = \text{Chip Rate}/\text{Data Rate}$.

Wireless Network Planning and Link Budget**Q1. What is wireless network planning?**

Answer: Wireless network planning is the process of designing a wireless system to provide required coverage, capacity and quality while efficiently using available spectrum and controlling interference.

Q2. What is a link budget?

Answer: A link budget is a calculation of the received signal power by accounting for transmitter power, antenna gains, path loss and other gains or losses in a wireless link.

Q3. Write the basic link-budget equation in dB form.

Answer: $P_r = P_t + G_t + G_r - L_{\text{total}}$, where P_r is received power, P_t is transmitted power, G_t and G_r are antenna gains, and L_{total} represents total losses.

Q4. Differentiate dB and dBm.

Answer: dB represents a ratio or gain/loss between powers. dBm represents an absolute power level referenced to 1 mW. For example, 0 dBm = 1 mW.

Q5. What is antenna gain?

Answer: Antenna gain indicates how effectively an antenna concentrates radiated or received power in a particular direction compared with a reference antenna. It is commonly expressed in dB or dBi.

Q6. What is path loss?

Answer: Path loss is the reduction in received signal power as a radio wave propagates from the transmitter to the receiver.

Q7. What is free-space path loss?

Answer: Free-space path loss is the attenuation experienced by a radio signal propagating through ideal free space without reflections, scattering or obstacles.

Q8. Write the FSPL formula for distance in km and frequency in MHz.

Answer: $FSPL(dB) = 32.44 + 20 \log_{10}(d_{km}) + 20 \log_{10}(f_{MHz})$.

Q9. What is receiver sensitivity?

Answer: Receiver sensitivity is the minimum received signal power required by a receiver to achieve the specified performance or quality.

Q10. Define link margin and write its formula.

Answer: Link margin is the difference between the actual received power and the minimum required receiver sensitivity. $Link\ Margin = P_r - P_{sensitivity}$.

Q11. What does a positive link margin indicate?

Answer: A positive link margin indicates that the received signal is stronger than the minimum required level, so the link has a power margin for reliable operation.

Power Spectrum**Q1. What is meant by power spectrum?**

Answer: A power spectrum shows how the power of a signal is distributed over frequency.

Q2. What is power spectral density (PSD)?

Answer: Power spectral density describes the amount of signal power present per unit frequency bandwidth.

Q3. What is the unit of power spectral density?

Answer: Power spectral density is commonly expressed in W/Hz or dBm/Hz.

Q4. Why is power spectrum analysis important in wireless network planning?

Answer: It helps determine how signal power is distributed over frequency, supports bandwidth planning, and helps identify interference and spectrum-usage problems.

PART B — 5-MARK QUESTIONS WITH ANSWERS

Multiple Access Techniques

Q1. Explain FDMA with a suitable diagram. Discuss its working, advantages and disadvantages.

Answer:

Definition: FDMA divides the available frequency spectrum into separate frequency bands and assigns each user a dedicated band.

Working: The total spectrum is divided into channels. Each user transmits continuously on an assigned frequency band. Guard bands are normally provided between adjacent channels to reduce interference.

Diagram: Frequency → | User 1 | Guard | User 2 | Guard | User 3 |

Advantages: simple implementation, continuous transmission, and no strict time-slot synchronization.

Disadvantages: guard bands waste spectrum, fixed allocation can be inefficient for uneven traffic, and adjacent-channel interference must be controlled.

Q2. Explain TDMA with a suitable diagram. Discuss time slots, guard time, synchronization, advantages and disadvantages.

Answer:

Definition: TDMA allows multiple users to share the same frequency channel by assigning different time slots.

Working: A repeating TDMA frame is divided into slots. Each user transmits during its assigned slot. Guard time is inserted between slots to prevent overlap.

Diagram: Time → | U1 | Guard | U2 | Guard | U3 | Guard | U4 |

Advantages: efficient sharing of a carrier, suitable for digital communication and flexible time-slot allocation.

Disadvantages: requires synchronization, guard time causes overhead, and timing errors can cause interference.

Q3. Explain CDMA with a suitable diagram. Explain spreading, chips, codes and the basic receiving process.

Answer:

Definition: CDMA allows users to transmit simultaneously using the same frequency band but with different spreading codes.

Working: The user's data is combined with a high-rate spreading code, producing a wider-band signal. The small elements of the spreading sequence are called chips. At the receiver, the desired user's code is used for correlation/despreading to recover the data.

Diagram: U1 + Code 1, U2 + Code 2, U3 + Code 3 → Same time + same frequency → Receiver → selected code → desired data.

Key point: Ideally, orthogonal or low-correlation codes allow users to be separated.

Q4. Explain the near-far problem in CDMA and how power control helps solve it.

Answer:

In CDMA, users may be at different distances from the base station. A nearby user's signal can arrive much stronger than a distant user's signal. The strong signal may interfere with the weak signal; this is called the **near-far problem**.

Solution: power control adjusts each mobile's transmit power so that signals arrive at appropriate levels at the base station. A nearby mobile generally needs less transmit power, while a distant mobile may need more. This reduces unnecessary interference and improves system capacity and reliability.

Q5. Compare FDMA, TDMA and CDMA with respect to signal separation, spectrum usage, synchronization, guard requirements, advantages, disadvantages and applications.

Answer:

FDMA: separates users by frequency; users occupy different frequency bands; guard bands are used; synchronization requirements are relatively low; simple but guard bands waste spectrum.

TDMA: separates users by time; users share the same carrier in different time slots; guard time is used; synchronization is important; efficient for digital systems but timing errors and overhead are concerns.

CDMA: separates users by codes; users can share the same time and frequency; spreading codes are used; synchronization and power control are important; high capacity and soft-handoff capability, but more complex and affected by the near-far problem.

Memory: FDMA → Frequency, TDMA → Time, CDMA → Code.

Spectral Efficiency

Q1. Define spectral efficiency and explain its significance. Derive/use $\eta = R/B$ with a suitable numerical example.

Answer:

Spectral efficiency measures the amount of information transmitted per unit bandwidth.

Formula: $\eta = R/B$, where R is data rate in bits/s and B is bandwidth in Hz.

Example: If R = 20 Mbps and B = 10 MHz, then $\eta = 20/10 = 2 \text{ bits/s/Hz}$.

Higher spectral efficiency means more information can be transmitted using the same spectrum, which is important because wireless spectrum is limited.

Q2. Explain the factors affecting FDMA spectral efficiency, including channel bandwidth and guard-band overhead.

Answer:

In FDMA, total spectrum is divided into frequency channels. If N users each have data rate R_u , the total data rate is approximately $N R_u$, and ideal efficiency can be expressed as $\eta = N R_u / B_{\text{total}}$.

Guard bands are placed between adjacent channels to reduce adjacent-channel interference. These guard bands consume spectrum without carrying useful information, so practical efficiency is lower than the ideal value. Fixed channel allocation can also cause poor spectrum utilization when traffic is uneven.

Q3. Explain TDMA spectral efficiency and the effect of guard time and synchronization overhead.

Answer:

In TDMA, multiple users share a common frequency channel by transmitting in separate time slots. If the channel provides total useful data rate R over bandwidth B , ideal spectral efficiency is $\eta = R/B$.

In practice, part of each frame is used for guard time, synchronization and control information. A simple time-efficiency factor is $T_{\text{useful}}/T_{\text{frame}}$. Therefore, practical efficiency is reduced by overhead.

For equal slots, each of N users gets approximately $1/N$ of the available transmission time.

Q4. Explain CDMA processing gain and its relationship with chip rate, data rate and spread bandwidth.

Answer:

Processing gain measures the amount by which the signal is spread relative to the original data rate.

Formula: $G_p = W/R = \text{Chip Rate}/\text{Data Rate}$.

For example, if chip rate = 1.2288 Mcps and data rate = 9.6 kbps, $G_p = 1.2288 \times 10^6 / 9.6 \times 10^3 = 128$.

A higher processing gain indicates greater spreading of the signal and contributes to the spread-spectrum characteristics of CDMA.

Q5. Solve a numerical problem involving spectral efficiency, total data rate and bandwidth for a multiple-access system.

Answer:

Example: An FDMA system has 5 users. Each user transmits at 2 Mbps and the total available bandwidth is 10 MHz.

Total data rate = $N R_u = 5 \times 2 = 10$ Mbps.

Spectral efficiency = $R/B = 10/10 = 1$ bit/s/Hz.

The same basic relation, $\eta = R/B$, is used for system-level spectral-efficiency calculations. Always check whether the problem includes guard-band, guard-time or other overhead that must be accounted for.

Wireless Network Planning and Link Budget

Q1. Explain wireless network planning and the role of link-budget calculations in reliable communication coverage.

Answer:

Wireless network planning designs the locations, frequencies, powers and other parameters of a wireless system to achieve required coverage, capacity and quality while controlling interference.

A link budget estimates received signal power by accounting for transmitter power, antenna gains and propagation/system losses. It helps determine whether the received signal is above receiver sensitivity and whether sufficient link margin exists.

Q2. Explain the link-budget equation, including transmitter power, antenna gains, path loss and other system losses.

Answer:

In dB/dBm form, the basic link-budget equation is:

$$P_r = P_t + G_t + G_r - L_{\text{total}}$$

P_t is transmitter power, G_t and G_r are antenna gains, and L_{total} includes path loss, cable loss, connector loss and other losses. Gains are added and losses are subtracted. The result is the estimated received power.

Q3. Explain dB and dBm and show how they are used in link-budget calculations.

Answer:

dB: a logarithmic measure of a power ratio or gain/loss. For a power ratio, $\text{dB} = 10 \log_{10}(P_2/P_1)$.

dBm: an absolute power level referenced to 1 mW: $P(\text{dBm}) = 10 \log_{10}(P(\text{mW})/1 \text{ mW})$.

In link budgets, transmitter power is often given in dBm, while antenna gains and losses are given in dB. Gains are added and losses are subtracted.

Q4. Explain free-space path loss and use the FSPL expression for a wireless link.

Answer:

Free-space path loss is the attenuation of a radio signal propagating through ideal free space. It increases with both distance and frequency.

For distance in km and frequency in MHz:

$$\text{FSPL}(\text{dB}) = 32.44 + 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{MHz}}).$$

Example: for $d = 1 \text{ km}$ and $f = 900 \text{ MHz}$, $\text{FSPL} \approx 91.52 \text{ dB}$.

Q5. Solve a complete link-budget numerical problem to determine received power and link margin.

Answer:

Given: $P_t = 20 \text{ dBm}$, $G_t = 5 \text{ dB}$, $G_r = 2 \text{ dB}$, $d = 1 \text{ km}$, $f = 900 \text{ MHz}$, other loss = 2 dB , receiver sensitivity = -75 dBm .

$\text{FSPL} \approx 91.52 \text{ dB}$.

$$P_r = 20 + 5 + 2 - 91.52 - 2 = -66.52 \text{ dBm}.$$

$$\text{Link margin} = P_r - P_{\text{sensitivity}} = -66.52 - (-75) = 8.48 \text{ dB}.$$

Since the margin is positive, the link has sufficient received power under the given assumptions.

Q6. Explain receiver sensitivity and link margin. How can you determine whether a wireless link is feasible?

Answer:

Receiver sensitivity is the minimum received power required for the receiver to meet its specified performance. Link margin is the difference between actual received power and receiver sensitivity:

$$\text{Link Margin} = P_r - P_{\text{sensitivity}}$$

If link margin is positive, the received signal is above the required minimum and the link is generally feasible. If it is negative, the received power is insufficient under the stated conditions.

Power Spectrum

Q1. Explain power spectrum/power spectral density and its importance in wireless network planning.

Answer:

A power spectrum describes how signal power is distributed over frequency. Power spectral density (PSD) expresses power per unit bandwidth, commonly in W/Hz or dBm/Hz.

In wireless planning, spectrum analysis helps determine occupied bandwidth, identify interference, evaluate spectrum usage and ensure that transmissions fit within allocated frequency bands.

Q2. Explain how bandwidth, transmitted power and spectral distribution are considered in power-spectrum calculations.

Answer:

Bandwidth determines the frequency range over which power is distributed. Total transmitted power is the integral/sum of power over the occupied spectrum, while PSD indicates how much power is present per unit bandwidth.

For a simple uniform distribution, average PSD can be estimated as total power divided by occupied bandwidth:
 $\text{PSD} \approx P/B$. In practical systems the spectrum may not be uniform, so the actual spectral shape must be considered.

Q3. Solve a suitable numerical problem involving power spectral density/power spectrum and bandwidth.

Answer:

Example: A transmitter radiates 1 W uniformly over a bandwidth of 2 MHz. Find the average PSD.

$$\text{PSD} = P/B = 1 \text{ W} / (2 \times 10^6 \text{ Hz}) = 5 \times 10^{-7} \text{ W/Hz}.$$

Thus the average power spectral density is $0.5 \mu\text{W/Hz}$. If a problem asks for dBm/Hz, first convert the total power to dBm and then account for the bandwidth in Hz.

UNIT II — QUICK FORMULA SHEET

- **Spectral Efficiency:** $\eta = R/B$
- **Total Data Rate:** $R_{\text{total}} = N \times R_u$
- **Processing Gain:** $G_p = W/R = \text{Chip Rate/Data Rate}$
- **Processing Gain in dB:** $G_{p,\text{dB}} = 10 \log_{10}(G_p)$
- **Link Budget:** $P_r = P_t + G_t + G_r - L_{\text{total}}$
- **FSPL:** $\text{FSPL}(\text{dB}) = 32.44 + 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{MHz}})$
- **Link Margin:** $M = P_r - P_{\text{sensitivity}}$
- **Average PSD (uniform spectrum):** $\text{PSD} \approx P/B$

Exam priority: Focus especially on FDMA/TDMA/CDMA comparison, spectral-efficiency numericals, CDMA processing gain, link-budget numericals, FSPL and link margin.